PROJECT REPORT

Project Title: AI-Powered Crop Recommendation System Using Machine Learning

A Project Report

Submitted in partial fulfilment of the requirements for the award of the degree of

BACHELORS OF TECHNOLOGY

In

Computer Science and Engineering

Submitted by :

Ridhima (25SCS1003001614)
Pragati Kumari (25SCS1003001660)
Anshika Shukla (25SCS1003004158)
Arpita Maurya (25SCS1003001604)
Ankita Sharma (25SCS1003004436)

Under the supervision of
Prof. Aditya Bangia

IILM University, Greater Noida, Uttar Pradesh

(November 2025)

CERTIFICATE

This is to certify that the Project Report entitled “AI-Powered Crop Recommendation System using Machine Learning" is a bonafide record of original work carried out by Ridhima (25SCS1003001614), Pragati Kumari (25SCS1003001660), Anshika Shukla (25SCS1003004158), Arpita Maurya (25SCS1003001604), and Ankita Sharma (25SCS1003004436) under my supervision in partial fulfillment of the requirements for the award of the degree of Bachelors of Technology in Computer Science and Engineering, IILM University, Greater Noida.

The work presented in this report has not been submitted elsewhere for any other degree or diploma to the best of my knowledge and belief.

Project Guide
Prof. Aditya Bangia
School of Computer Science and Engineering
IILM University, Greater Noida

ACKNOWLEDGEMENT

We express our sincere gratitude to Prof. Aditya Bangia, our project guide, for his invaluable guidance, support, and encouragement throughout the development of this project. His consistent feedback, technical insights, and motivation have been instrumental in shaping the direction and successful completion of this project.

We also extend our heartfelt thanks to the School of Computer Science and Engineering, IILM University, Greater Noida, for providing the essential facilities, resources, and a conducive learning environment that enabled us to carry out this project effectively.

We would like to acknowledge the support of our classmates and peers, whose constructive suggestions and collaborative discussions helped us refine our ideas and strengthen our approach.

Finally, we express our deepest gratitude to our families, whose constant encouragement, patience, and belief in our abilities have been our greatest source of motivation throughout this journey.

CONTENTS

1. INTRODUCTION

2. RELATED WORKS

3. METHODOLOGY

4. IMPLEMENTATION

5. RESULTS AND DISCUSSION

6. FUTURE SCOPE

7. CONCLUSION

8. REFERENCES

1. INTRODUCTION

1.1 Introduction

Agriculture remains the fundamental backbone of the global economy, yet modern farming faces complex challenges. Farmers traditionally rely on generational knowledge, intuition, or trial-and-error to select crops, methods which are often inefficient and fail to adapt to rapidly changing environmental conditions. This Crop Recommendation System leverages data-driven techniques, specifically Artificial Intelligence (AI) and Machine Learning (ML), to provide scientific guidance to farmers.

The system processes various environmental and physiochemical data points including soil nutrients (Nitrogen, Phosphorus, Potassium), pH, temperature, humidity, and rainfall to predict and suggest the single most suitable crop for a given piece of land. By replacing guesswork with predictive analytics, this solution optimizes resource utilization, reduces risks, and promotes sustainable farming practices, thereby increasing both yield and profitability for the farmer.

1.2 Problem Statement

Farmers frequently struggle with the critical decision of crop selection due to a significant lack of scientific and contextual guidance. Relying on traditional methods is increasingly risky, especially with the accelerating impact of climate change and irregular rainfall patterns.

The core challenges addressed by this project are:

- **Low Yield and Profitability**: Unplanned or unsuitable crop selection leads to wasted resources and poor harvests.

- **Resource Inefficiency**: Excessive use of fertilizers and water to compensate for poor soil-crop match, leading to higher costs.

- **Environmental Degradation**: Unsustainable farming practices resulting from non-optimal crop choices, contributing to soil nutrient depletion.

- **Climatic Uncertainty**: The inability of traditional methods to account for real-time variations in local weather and climate data.

This project aims to solve this problem by developing a reliable, high-accuracy ML model to provide farmers with data-backed, actionable crop recommendations.

2. RELATED WORKS

The development of the Crop Recommendation System builds upon various established and emerging approaches:

- **Traditional Approaches**: These include manual soil testing, laboratory analysis, and personalized crop advisory provided by agricultural experts. While accurate, these methods are costly, time-consuming, and not scalable.

- **Rule-Based Systems**: Early expert systems stems used predefined 'if-then' rules (e.g., "IF soil_type=sandy AND rainfall > X THEN recommend=Millet"). These lack flexibility and cannot handle the non-linear complexity of environmental data.

- **Machine Learning Models**: Extensive research has been conducted using algorithms like Decision Tree (DT), K-Nearest Neighbors (KNN), Naïve Bayes, and Support Vector Machines (SVM) for crop prediction, demonstrating significant improvements over rule-based systems.

- **Deep Learning Approaches**: More complex studies involve Neural Networks and Ensemble Methods to push prediction accuracy beyond 95%, often integrating image recognition for disease detection alongside crop recommendations.

- **IoT and Sensor-Based Systems**: Recent advancements integrate the recommendation models with Internet of Things (IoT) sensors to monitor soil moisture, NPK levels, and microclimate factors in real-time, providing immediate and dynamic crop suggestions.

3. METHODOLOGY

The development of the Crop Recommendation System followed a structured, data-driven approach based on the standard Supervised Machine Learning pipeline. The primary goal was to systematically compare classification algorithms and select the most accurate and robust model for predicting optimal crop choices.

3.1 System Architecture and Process Flow

The system architecture can be broken down into three main phases: Data Management, Model Training, and Prediction/Recommendation. The overall process flow is summarized in the following diagram:

Phase 1: Data Management - This phase focuses on acquiring the raw data and transforming it into a clean, structured format suitable for machine learning algorithms. It includes:

1. Data Acquisition: Gathering the dataset containing seven essential features (N, P, K, pH, Temperature, Humidity, Rainfall) and the target crop label.

2. Data Preprocessing: Cleaning the data by handling null values, treating outliers, and performing necessary transformations, such as converting the categorical crop labels into numerical classes (Label Encoding).

3. Feature Scaling: Applying Min-Max Scaling to the numerical features (e.g., N, P, K, Rainfall) to ensure that no single feature dominates the model training solely based on its magnitude.

4. Data Splitting: Partitioning the prepared dataset into an 80% Training Set and a 20% Testing Set to validate the model's performance on unseen data.

<u>Phase 2</u>: Model Training and Selection - This phase involves the core machine learning activities:

1. Algorithm Selection: Four robust classification algorithms were chosen for comparative analysis: Decision Tree (DT), K-Nearest Neighbors (KNN), Support Vector Machine (SVM), and Random Forest (RF).

2. Training and Tuning: Each model was trained using the Training Set. Hyperparameter tuning was performed, particularly for the Random Forest classifier (adjusting the number of estimators, or n estimators), to maximize its predictive accuracy.

<u>Phase 3</u>: Evaluation and Recommendation - The final phase validates the model's success and generates the output:

1. Model Evaluation: Using the Testing Set, the performance of all models was measured using key metrics (Accuracy, Precision, Recall, and F1-Score).

2. Model Selection: Based on the highest accuracy and overall robustness, the Random Forest model was chosen as the definitive recommendation engine.

3. Prediction: The finalized model is deployed to accept new user inputs (e.g., current soil test results) and predict the optimal crop.

3.2 Machine Learning Algorithms

The project focused on a comparative study to ensure the optimal choice of the predictive engine:

- **Random Forest (RF)**: An Ensemble Learning method that constructs multiple Decision Trees during training. It combines the predictions of these individual trees to output the most frequent prediction. This approach is highly effective in handling non-linear data, reducing overfitting, and providing superior accuracy.

- **Support Vector Machine (SVM)**: A model that finds the optimal hyperplane that distinctly separates data points into different classes (crops) in a high-dimensional space.

- **K-Nearest Neighbors (KNN)**: A simple, non-parametric, instance-based algorithm that classifies a new data point based on the majority class among its K nearest neighbors in the feature space.

- **Decision Tree (DT)**: A foundational model that uses a tree-like structure of decisions (rules) and their possible consequences to classify the data.

4. IMPLEMENTATION

The Crop Recommendation System was implemented using the Python programming language, capitalizing on its rich ecosystem for data science and machine learning.

4.1 Tools and Technologies

Component	Technology	Purpose
Programming Language	**Python 3.x**	Primary language for development and scripting.
Development Environment	**Jupyter Notebook / Google Colab**	Interactive environment for data exploration and model development.
Core ML Library	**Scikit-learn (Sklearn)**	Used for all model implementations, training, testing, and evaluation.
Data Handling	**Pandas, NumPy**	Data ingestion, manipulation, numerical computation, and feature processing.
Visualization	**Matplotlib, Seaborn**	Generating plots for data exploration and model comparison (e.g., the Algorithm Accuracy Bar Chart).

4.2 Code Implementation Summary

The implementation followed a clear sequence of steps executed in the Python environment:

1. Data Loading and Initial Exploration: The project began by loading the Kaggle dataset into a Pandas DataFrame. Initial steps included checking the data shape, data types of columns, and performing a quick statistical summary (mean, median, standard deviation) to understand the distribution of features like NPK and Rainfall.

2. Feature and Target Separation: The dataset was divided into independent variables (X) comprising the environmental features and the dependent variable (Y) containing the 'label' (crop name).

3. Data Preprocessing and Scaling: The target variable Y was encoded numerically using LabelEncoder. The feature matrix X was scaled using MinMaxScaler to ensure all inputs ranged from 0 to 1, stabilizing the model training process.

4. Model Training and Selection: The processed data was split using train test split. The Random Forest Classifier was instantiated with optimized hyperparameters and trained on the training data.

```
# Example of the core Random Forest implementation logic
from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score

# Assuming X_scaled and Y_encoded are prepared
✗ _train, X_test, Y_train, Y_test = train_test_split (X_scaled, Y_encoded, test_size=0.2,
random_state=42)

# Instantiate the model
rf_model = RandomForestClassifier (n_estimators=100, random_state=42)

# Train the model
rf model.fit (X train, Y train)

# Predict and evaluate
predictions = rf model.predict (X_test)
accuracy = accuracy_score (Y_test, predictions)

# . . . further evaluation with Confusion Matrix and Classification Report
```

5. Model Evaluation and Prediction: The model's performance was rigorously tested against the Testing Set. The resulting predictions were compared with the actual crop labels (Y_{test}) to generate the final accuracy score and detailed metrics. The high accuracy (~ 98%) confirmed the Random Forest model as the production-ready classifier.

5. RESULTS AND DISCUSSION

The rigorous evaluation of the implemented Machine Learning models provided clear, measurable results demonstrating the project's success in developing a high-accuracy crop recommendation engine.

5.1 Performance Metrics Evaluation

The models were evaluated on the 20% Testing Set using standard classification metrics: Accuracy, Precision, Recall, and F1-Score.

Metric	Definition	Importance for Crop Recommendation
Accuracy	Ratio of correctly predicted observations to total observations.	Overall measure of the system's reliability.
Precision	Measure of how many of the recommended crops were actually correct.	Minimizes recommending the *wrong* crop (false positives).
Recall	Measure of how many of the truly suitable crops were correctly identified.	Maximizes the chance of finding the *best* possible crop (minimizes false negatives).
F1-Score	Harmonic mean of Precision and Recall.	Provides a single, balanced score for model performance.

5.2 Comparative Model Performance

The models were trained and tested on the Kaggle Crop Recommendation dataset, which includes 22 different crops and 7 key input features (N, P, K, pH, Temperature, Humidity, Rainfall).

Algorithm	Achieved Accuracy	Performance Characteristics
Decision Tree (DT)	~92%	High interpretability, prone to overfitting.
K-Nearest Neighbors (KNN)	~94%	Simple and effective, sensitive to irrelevant features.
Support Vector Machine (SVM)	~95%	Good for complex boundaries, but computationally expensive.
Random Forest (RF)	**~98%**	**Highest accuracy, robust against noise, industry standard.**

5.3 Discussion of Results

A. <u>The Superiority of Random Forest</u>

The Random Forest (RF) algorithm consistently outperformed all other models, achieving an accuracy of approximately 98%. This exceptional performance is attributed to the inherent design of ensemble methods:

1. Bias-Variance Tradeoff: Random Forest mitigates the high variance (overfitting) typically seen in a single Decision Tree by averaging the predictions of many trees. This leads to a more generalized and reliable model.

2. Robustness: By introducing randomness both in the data samples (bootstrapping) and in the features considered for each split, the model became highly robust to noise and slight variations in the agricultural dataset.

This high level of predictive performance and technical robustness strongly justified the selection of the Random Forest model as the final recommender engine for deployment.

B. <u>Practical Implications of the Result</u>

The achieved ~98% accuracy translates directly into significant practical benefits for the agricultural sector:

- Risk Mitigation: With a near-perfect prediction rate, farmers can drastically reduce the risk associated with crop failure due as a result of unsuitable soil or weather conditions.

- Resource Optimization: By recommending the specific crop best suited for the available NPK, pH, and rainfall, the system minimizes the unnecessary use of of fertilizers, f water, and labor, promoting sustainable farming and higher profit margins.

- Actionable Intelligence: The model provides clear, data-backed guidance, replacing subjective, traditional methods with objective, scientific analysis.

C. Feature Importance

Further analysis of the Random Forest model revealed the relative importance of the input features in driving the final recommendation. The features identified as most critical were:

1. Nitrogen (N) Content: Essential for plant growth and development.

2. Potassium (K) Content: Crucial for water regulation, disease resistance, and fruit quality.

3. Rainfall: The most volatile climatic factor, highly influencing crop selection (e.g., rice requires high rainfall; cotton requires less).

The strong reliance on these physiochemical and climatic factors validates the model's ability to truly understand the complex multivariate relationships in agricultural science.

6. FUTURE SCOPE

The success of the current Crop Recommendation System-validated by the high accuracy of the Random Forest model-lays a strong foundation for future expansion. The next phases of development will focus on integrating real-time data, enhancing user interaction, and evolving the predictive capabilities of the system.

6.1 Real-Time Data Integration with IoT

Currently, the model relies on historical and static dataset parameters. The most significant future enhancement involves moving to a dynamic system through the integration of the Internet of Things (IoT):

- Sensor Connectivity: Deploying low-cost IoT soil sensors to capture real-time, hyperlocal data points for soil moisture, temperature, and ambient humidity.

- Dynamic Recommendations: Feeding this live sensor data directly into the trained Random Forest model. This will allow the system to provide instant and dynamic recommendations, advising farmers to switch crops or make mid-cycle adjustments based on current environmental shifts.

6.2 Advanced Predictive Capabilities (Yield and Profitability)

Expanding the scope beyond mere classification (what crop to plant) to prediction (how much will it yield and how much profit):

- Yield Forecasting: Developing a regression model (e.g., using Deep Neural Networks or advanced ensemble methods) that uses the recommended crop and the corresponding environmental parameters to predict the expected harvest yield in metric tons or kilograms per acre.

- Profitability Analysis: Integrating current market price data (fetched via secure APIs) to estimate the potential profit margin for the recommended crop, allowing farmers to make financially optimal decisions.

6.3 Deployment and User Experience

To maximize the impact on farmer livelihoods, the system requires an accessible and intuitive user interface:

- Mobile Application Development: Creating a dedicated mobile application (for Android and iOS) that allows farmers to input their location and soil test results easily, or automatically connect to their IoT sensors. The application would deliver the recommendation instantly, complete with planting quidelines.

- Explainable AI (XAI): Implementing a module to provide transparent justification for the model's recommendation (e.g., "We recommend rice because your NPK levels are high, and the rainfall in your area is adequate, but avoid maize due to low pH."). This builds trust and provides valuable agricultural knowledge to the user.

6.4 Model and Data Enhancement

Continuous improvement of the core AI engine is necessary:

- Deep Learning Exploration: Replacing the Random Forest classifier with more complex Deep Learning architectures, such as Convolutional Neural Networks (CNNs) for spatial data analysis (if satellite imagery is included) or Long Short-Term Memory (LSTM) Networks for time-series forecasting of optimal planting windows.

- Handling Imbalance and Rare Crops: Implementing advanced techniques (like SMOTE) to ensure the model remains highly accurate even for less common or regionally specific crops, preventing bias towards highly represented crops.

Overall, the project has immense potential to evolve into a complete AI-Powered Decision Support Tool for precision agriculture, significantly boosting efficiency and sustainability across various farming communities.

7. CONCLUSION

The Crop Recommendation System successfully achieves its primary objective of developing a high-accuracy, data-driven solution for enhancing agricultural decision-making. By applying Supervised Machine Learning techniques to critical environmental and physiochemical parameters (N, P, K, pH, Temperature, Humidity, and Rainfall), this project provides a reliable alternative to traditional, intuition-based farming methods. The systematic methodology, including a rigorous comparative analysis of four classification algorithms (DT, KNN, SVM, and RF), culminated in the selection of the Random Forest model. This model demonstrated exceptional predictive capability, achieving a verified accuracy of approximately 98% on unseen test data. This result not only validates the model's technical success but also confirms its practical utility in real-world agricultural scenarios.

This system represents a crucial step towards modernizing agriculture. By providing farmers with precise, scientific recommendations, the project contributes significantly to:

- Increased Productivity and Yield: By ensuring the chosen crop is perfectly suited to the land's conditions.

- Sustainable Resource Management: By optimizing the input of fertilizers and water.

- Reduced Financial Risk: By mitigating crop failure due to unsuitable planting decisions.

In essence, the Crop Recommendation System is a robust, reliable tool that is poised to contribute to enhanced food security and improved farmer livelihoods across various climatic zones.

8. REFERENCES

This project utilized several key resources, including foundational academic literature, industry-standard datasets, and development tools, which are formally cited below:

8.1 Data and Software Resources

1. [Dataset Source] Kaggle. Crop Recommendation Dataset. Contains the physiochemical (N, P, K, pH) and climatic (Temperature, Humidity, Rainfall) data used for model training and testing.

2. [Software] Python Software Foundation. Python 3.x. The primary programming language used for all implementation.

3. [Library] Scikit-learn (Sklearn). The core Machine Learning library utilized for algorithm implementation (Random Forest, SVM, KNN, DT), data preprocessing, and model evaluation.

4. [Library] Pandas & NumPy. Essential libraries used for data handling, cleaning, and numerical operations.

8.2 Academic and Foundational Resources

1. [Research Paper 1] Crop Recommendation System using Machine Learning Approaches (General overview and methodology for ML application in agriculture).

2. [Research Paper 2] Evaluation of Classification Algorithms for Crop Recommendation based on Environmental Parameters.

3. [Academic Source] Discussion on Ensemble Learning Methods in Predictive Agriculture (Focus on Random Forest performance and robustness).

8.3 [Support] Google Gemini (Used for debugging, code refinement, and technical assistance during the implementation phase).